



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



Explainable AI in Finance: Beyond the Code

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Funded under the Marie Skłodowska-Curie Actions (MSCA), the network brings together academic institutions, financial companies, technology providers and policy organizations to jointly train the next generation of researchers in digital finance



About Me

MSCA PHD FELLOW (STATISTICS)

University of Naples Federico II

Current research: Sustainable finance, explainable AI

LAUREA MAGISTRALE SCIENZE ECONOMICHE

Università degli studi Roma Tre, Italy

Sorbonne Université, France

Thesis: Blockchain for Solidarity Finance and Local
Development: Case of Sarafu Community Currency in
Kenya

BS DEVELOPMENT ECONOMICS

University of the Philippines Los Baños

Thesis: Efficient Market Hypothesis: Evidence
from the Bitcoin Market



Decisions that affect livelihoods
are now run by AI...so what?

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Decisions that affect livelihoods are now run by AI...so what?

Fairness judgement is outsourced

AI shifts decision-making from human interaction (banker-client relationship) to "automated discretion," which can alter the "social contract" of fairness.

"Blackboxes" act as gatekeeper

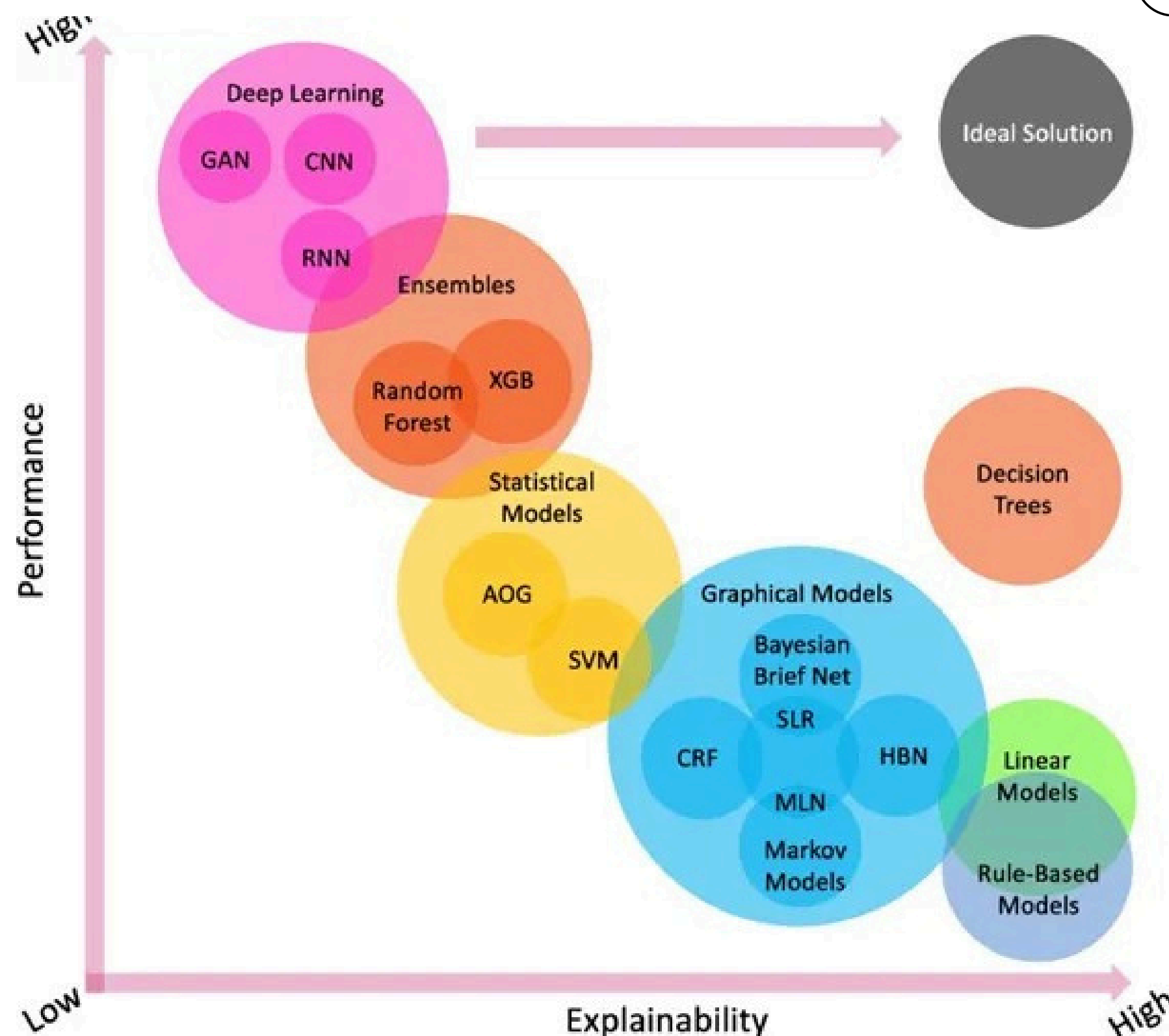
When the AI's logic is opaque, it challenges the individual's right to contest institutional decisions.

Some XAI Basics

What are blackbox models?

MODELS WHOSE INTERNAL LOGIC IS NOT DIRECTLY INTERPRETABLE BY HUMANS

- "White-box": linear regression, decision trees vs "black-box" neural networks or ensemble methods
- Literature in finance (see Zetzsche et al., 2020; Kumar, 2017; Cernevicene et al., 2024; Ogunniyi, et al., 2025)



Why choose black-boxes?

"THE CURSE & GIFT OF DEPTH"



Optimization

Models learn by minimizing the loss through gradient descent.



Non-linearity

Activation functions (e.g., ReLU, sigmoid) allow complex decision boundaries.



Overparameterization and emergent structure

Enables pattern discovery but hides reasoning

Deep layers encode abstract manifolds = statistically efficient, but cognitively opaque

Explainable AI

METHODS THAT MAKE BEHAVIOR, DECISIONS,
AND INTERNAL LOGIC UNDERSTANDABLE
TO HUMANS



Scope

Global - understand the overall behavior of the model

e.g., feature importance, partial dependence plots (PDP)

Local - explain a single prediction
e.g., LIME, SHAP, counterfactuals

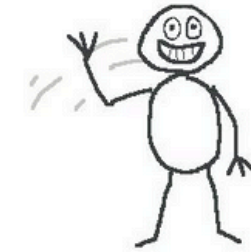


Dependence

Model-specific - works with only certain algorithms (e.g., coefficients in linear / logistic regression, tree visualizations)

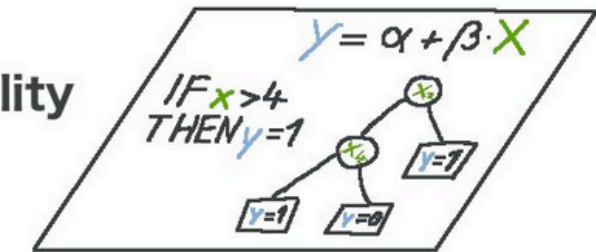
Model-agnostic - works with any model by proving input-output behavior (e.g., SHAP, LIME)

Humans



↑ inform

Interpretability
Methods



↑ extract

Black Box
Model



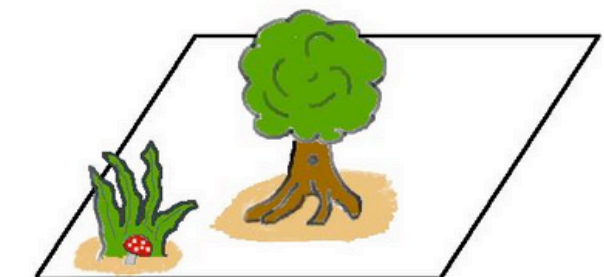
↑ learn

Data

X_1	X_2	X_3	...	X_n
1	5	2	...	0.1
2	3	1	...	0.5
...	...	...	...	...
n	...	...	...	...

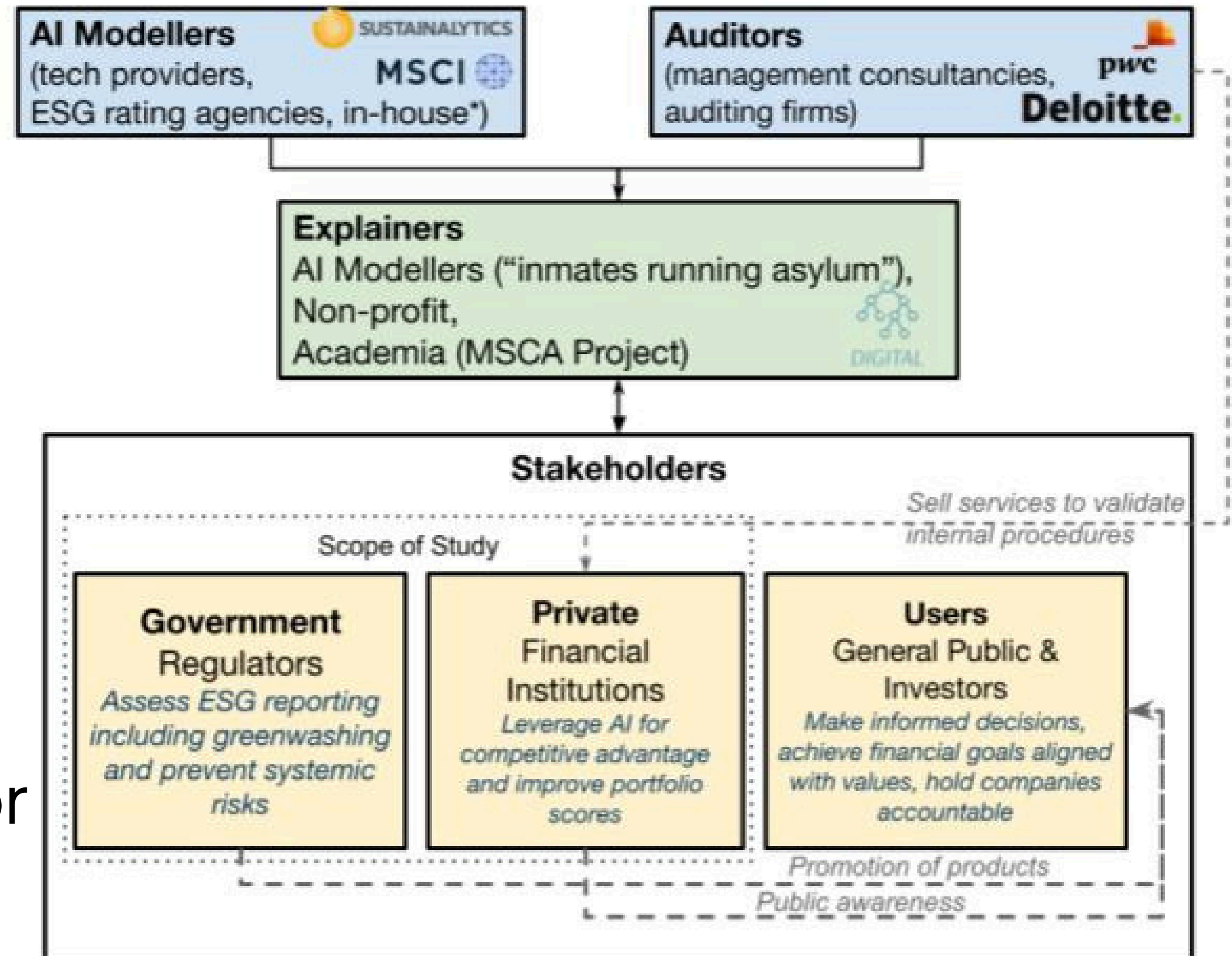
↑ capture

World



Main issues

- State of XAI: "inmates running the asylum" (Miller, 2019)
- In sustainable finance, competing interests
- Incumbents in the field (Fligstein, 2012)
- Performative XAI; only for legal requirement?



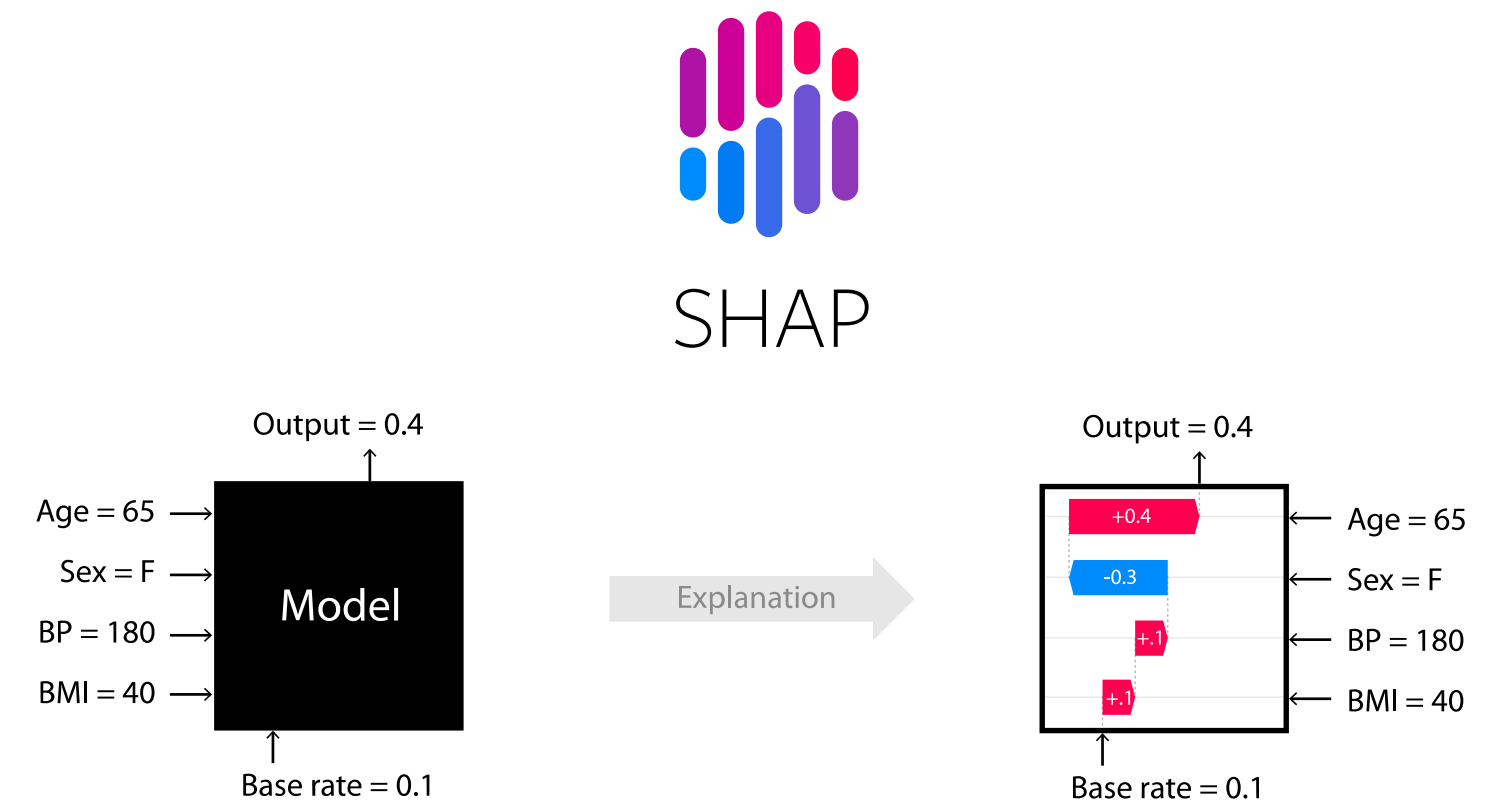
Research Problem

WHAT I WANT TO SOLVE

XAI methods in finance are focused on technical users (SHAP, LIME) like "inmates running the asylum" (Miller et al., 2017)

Incomprehensible AI leads to low AI trust and adoption (Bernardo & Seva, 2023)
Overtrust in incapable AI (Ehsan et al., 2021)

Literature suggests audience-dependent explanations (Misheva & Osterrieder, 2023; Schmude et al., 2024)



"[On shapley values] it is not
intuitive enough for a user to
explain it." — B2B Sales Officer, AI Vendor (Europe)

“The business would not understand it. They were so used to
the regression coefficients... if you increase this variable,
therefore the value will increase.”

— Data scientist, Bank (Asia)

Roles in Interpretable AI

Molnar (2025) and author



(Affected)
User

e.g.
customers/
public



Operator

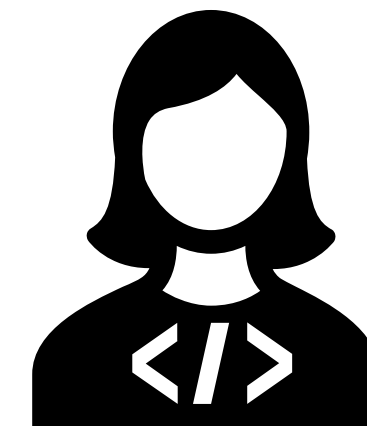
aka
business
users
(marketing,
compliance,
etc.)

Executor

makes
decisions
based on
AI

Manager

responsible
for the
project



Developer

creator

Validator

checks
model's
outputs



Regulator



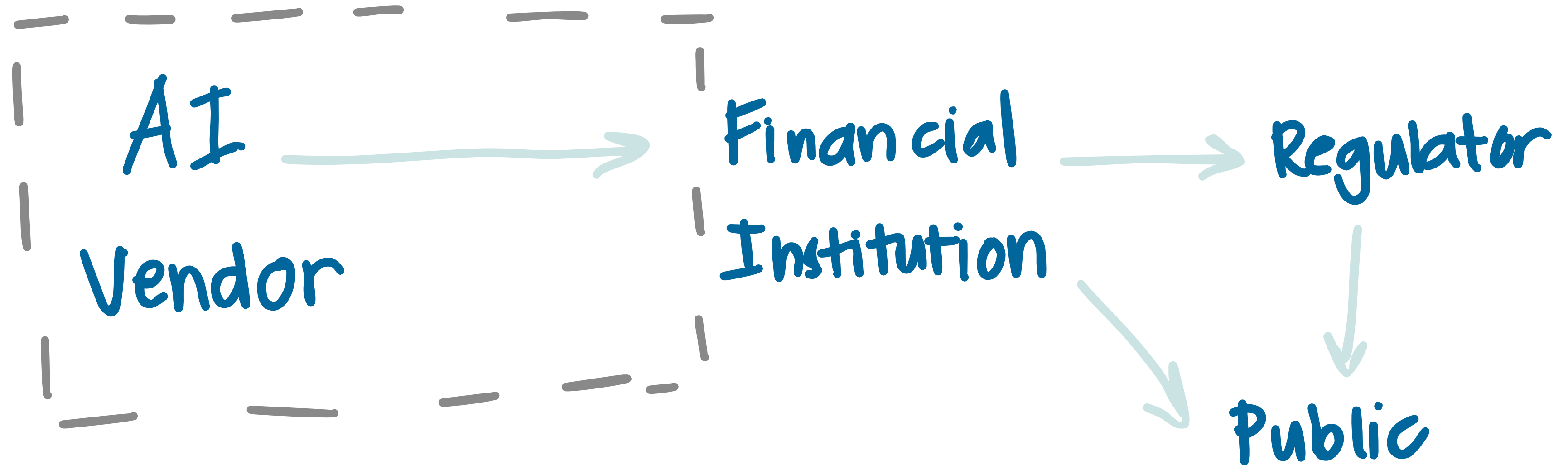
Conduct

consumer
protection,
fair trade

Prudential

systemic
financial
risk (ECB)

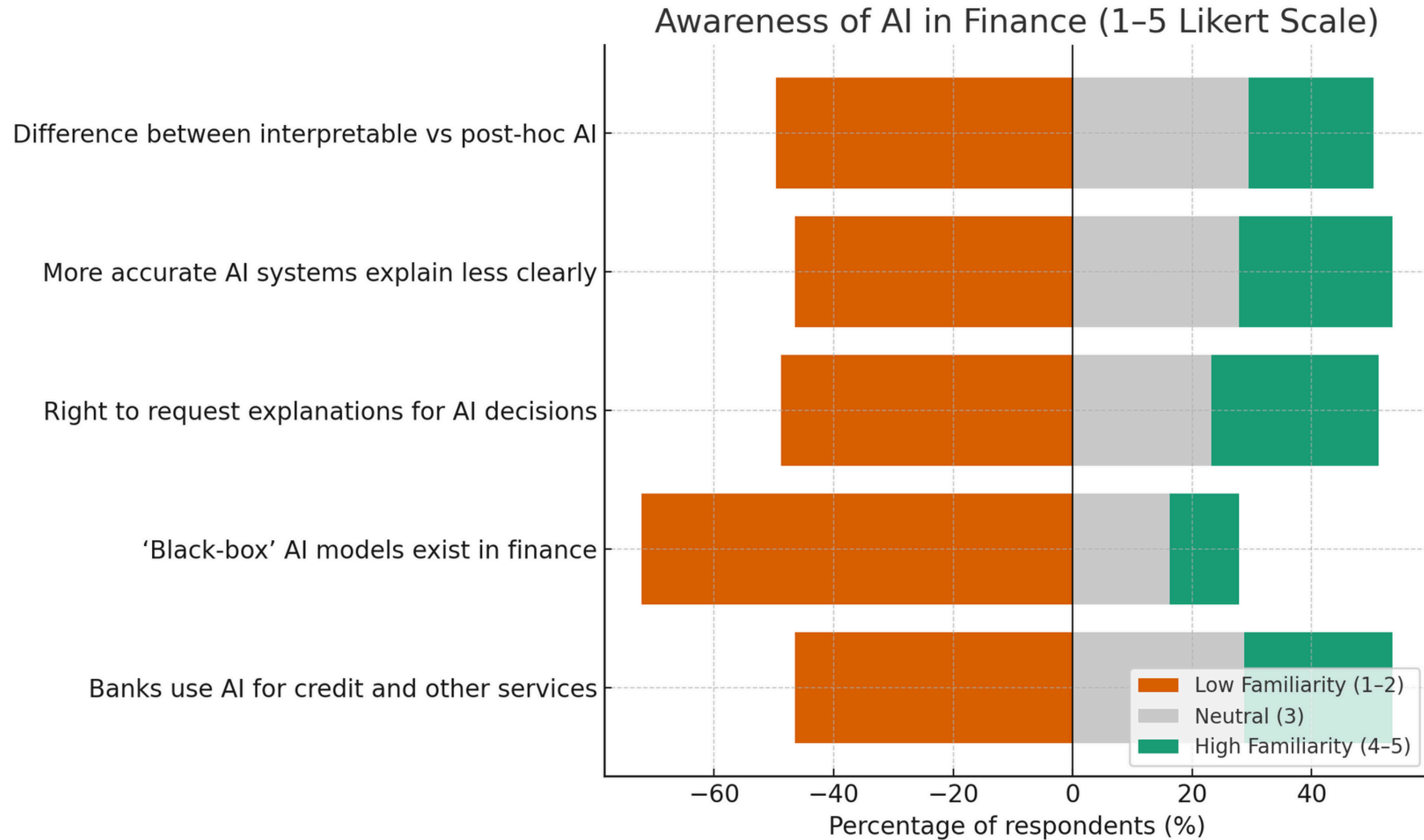
In-house vs proprietary AI tools



→ Direction of
XAI accountability

Preliminary Findings

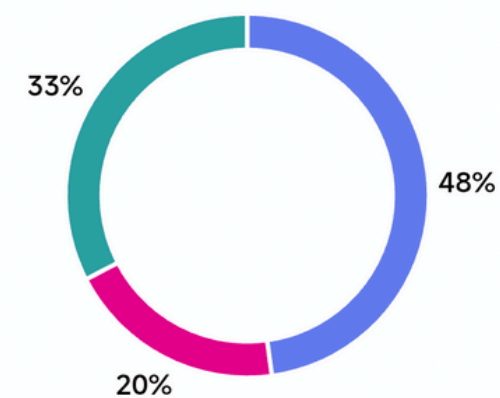
Public / App Users



8. Per una qualsiasi delle attività che hai selezionato, l'app ti ha fornito una spiegazione?

[More details](#)

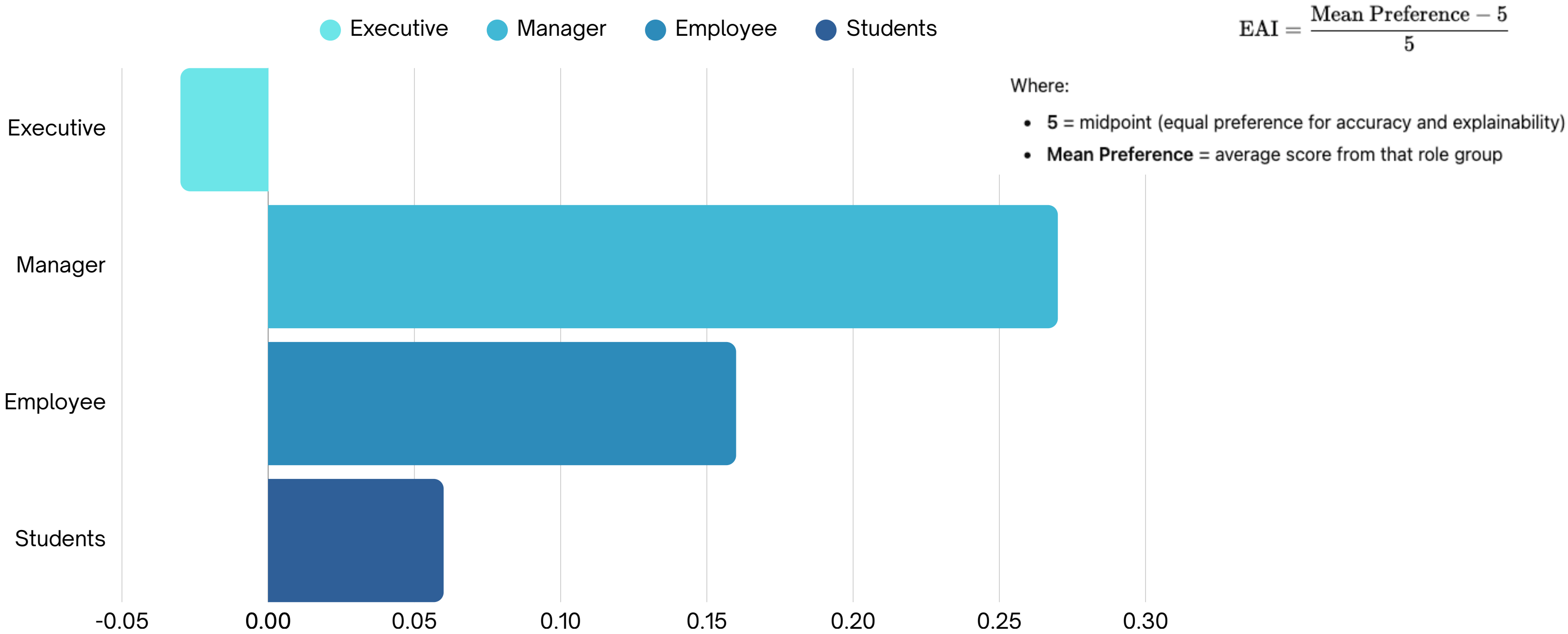
● Sì – Ho ricevuto una spiegazione	22
● No – Non ho ricevuto alcuna spiegazione	9
● Non lo so / Non ne sono sicuro/a	15



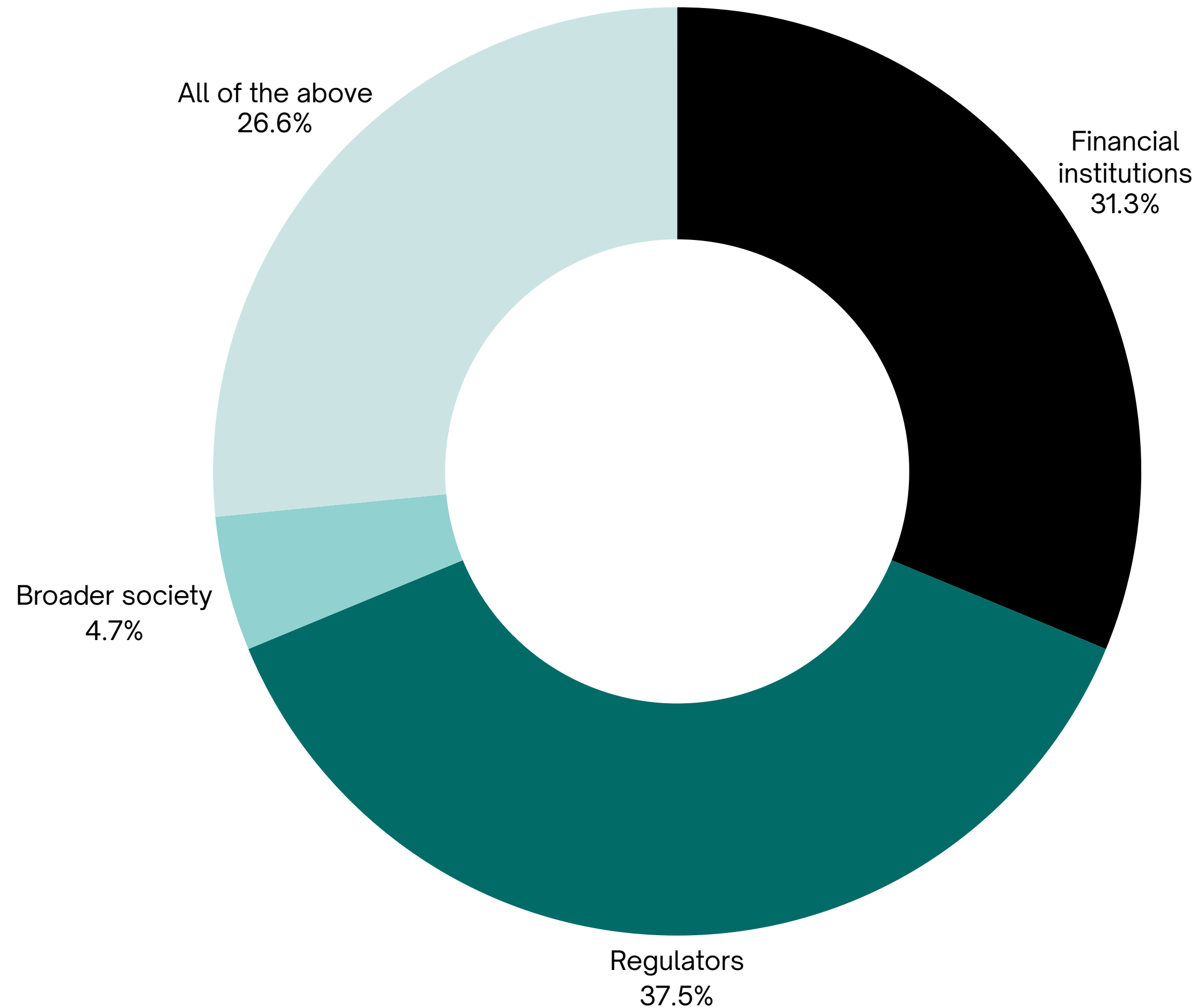
6 out 10 people received an explanation. Not all those who used financial apps with automated decisions received an explanation. Those who did often described it as unclear, incomplete, or generic.

Executives value accuracy over explainability.

Participants rated their ideal AI system from 0 (accuracy-focused) to 10 (explainability-focused).

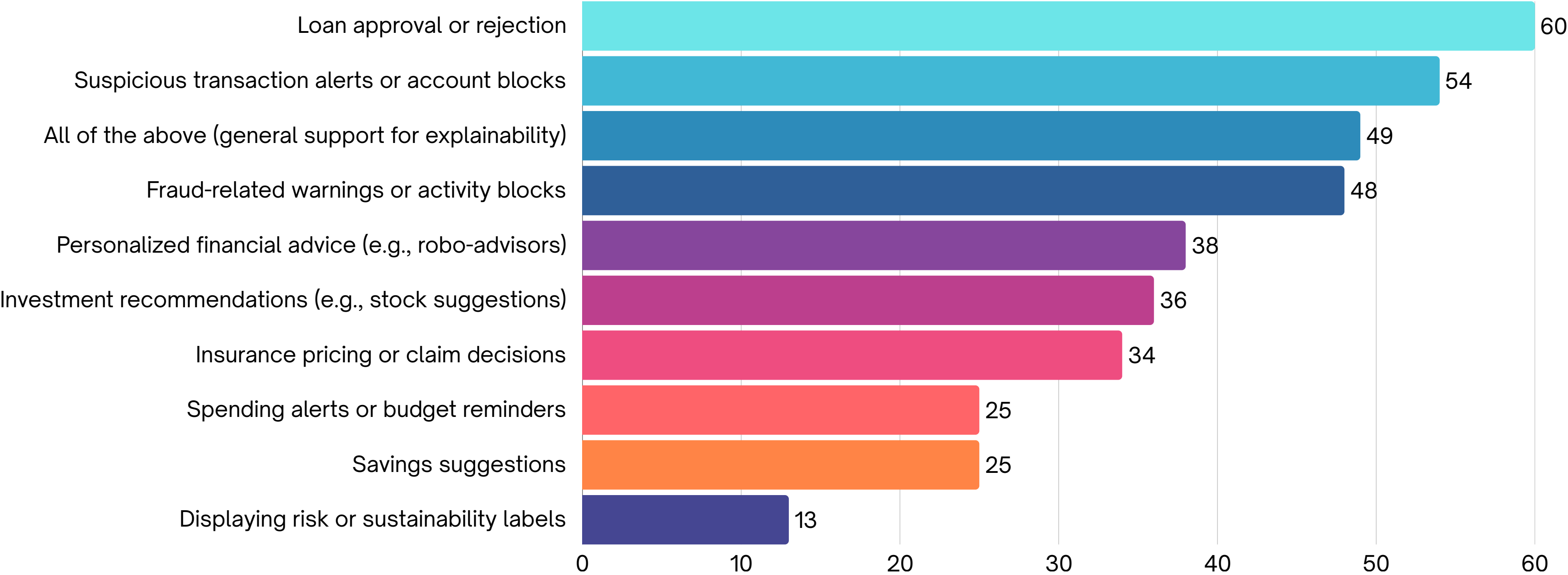


Who should be responsible for ensuring AI transparency in financial apps?



Which financial transactions should have mandatory explanations?

The strongest demand for explainability appears in high-impact, decision-critical areas, especially credit scoring (loan approvals) and fraud management.



Next steps

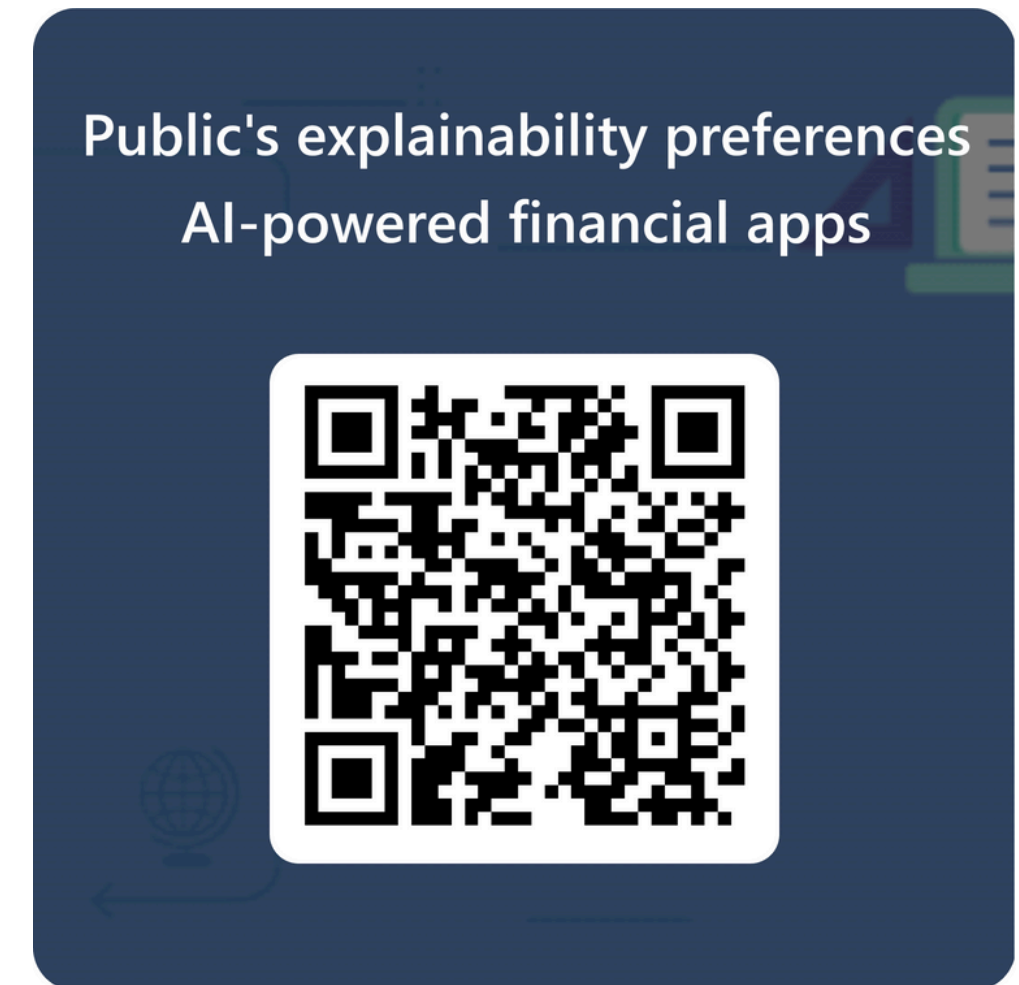
January 12

Italian version:

<https://forms.cloud.microsoft/e/gQwC14dS1i>



English version:
<https://forms.cloud.microsoft/e/hXMAtdXKQq>



January 15

MSCA Digital Finance Research
Seminar online hosted in Romania

January 30

Deadline for 20 respondents per
student

Thank you!